

ArtStation blog tutorial - sliders

My background is in fashion design. Before I ever opened Marvelous Designer I spent years working with real fabric, learning how raw denim resists when you pull it on the bias, and why a bias-cut dress moves so differently from a tailored jacket.

When I wanted to go beyond presets and start tuning my fabrics properly, I went looking for a clear explanation of what each physical property slider actually does. I couldn't find one. So I put this together myself, by handling real fabric samples and matching what I felt to what the simulation was doing. I reached out to the Marvelous Designer team to fact-check my findings, and they were kind enough to share their internal documentation with me which I've used to finalise this tutorial.

I use two fabrics throughout this tutorial. Raw denim and a pleated knit, because they sit at opposite ends of almost every property. I picked them up, pulled them, bunched them, compressed them, and dropped them. Then I matched what I felt to what the sliders were doing.

Let's dive into it!

STRETCH - *Resistance to being pulled*

Raw denim barely moves when you pull it on the straight grain. On the bias it gives a little, but it's still fighting you. High stretch resistance means the simulation won't let the fabric deform much when the character moves.

Knit gives immediately and springs back when pulled in any direction. Low stretch resistance. In simulation this fabric moves with the body, hugs the form, stretches over joints without fighting. Think leggings, a bodysuit, a tight undershirt on a character.



Denim

Pleated knit



Stretch 0



Stretch 100

When you're dressing a character in tight-fitting clothing, stretch is the slider that determines whether it looks like it's actually hugging the body or whether it looks like a human wearing wrapping paper.

SHEAR - Resistance to diagonal distortion

When you hold the denim at its diagonal corners and gently pull them apart, almost nothing moves. The fabric holds its rectangular shape. The threads lock in place. This is why tailored garments in stiff fabrics hold clean geometric shapes.

The same grip on the knit and it shifts and distorts immediately. The structure of the knit allows the loops to move relative to each other. This is also what happens

with bias-cut fabric. When you cut a woven fabric on the bias you are essentially making it behave as if it has low shear resistance.



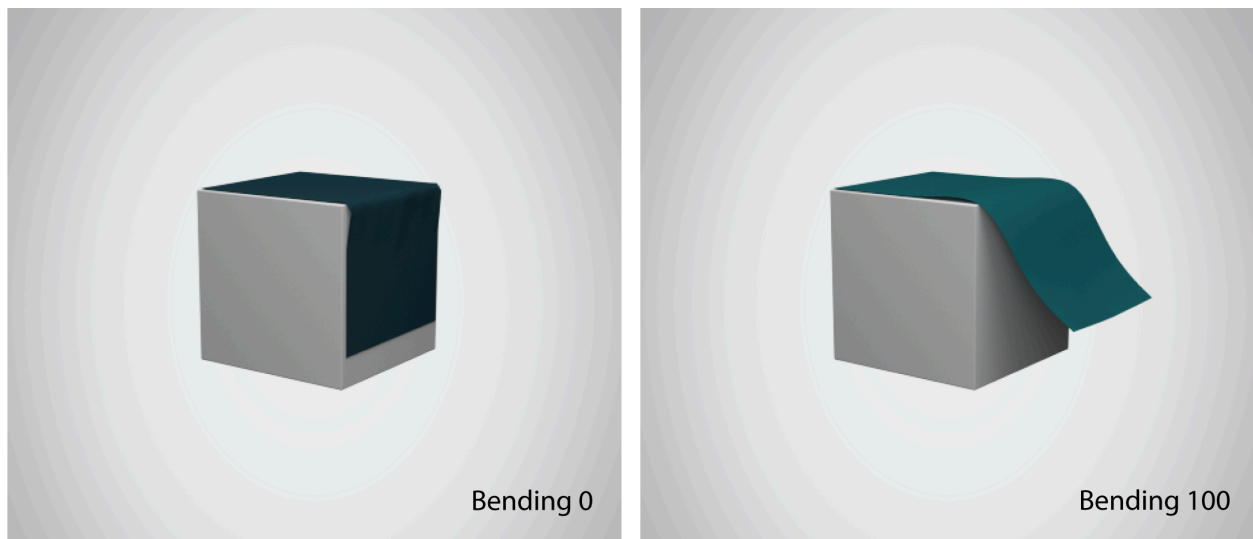
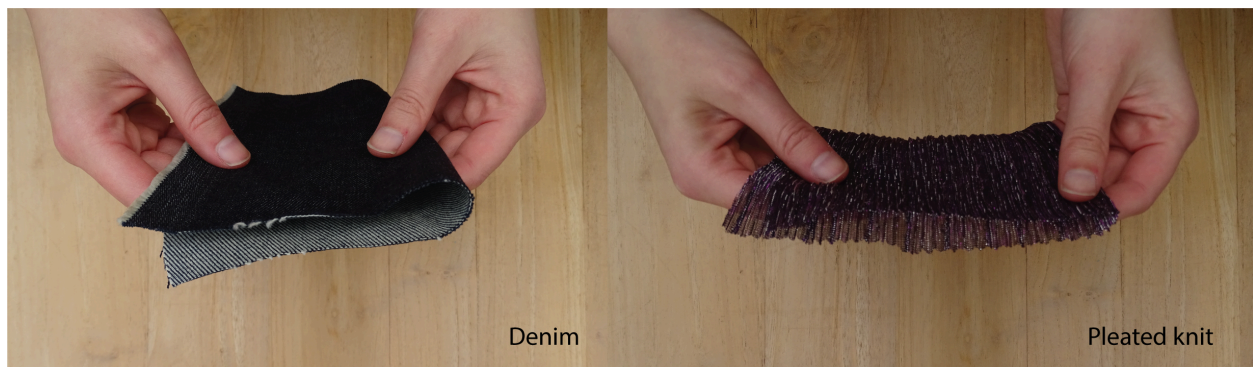
Shear is why a bias-cut gown moves so fluidly and why a tailored wool jacket holds its square shoulders. In simulation, this is the slider that controls whether your fabric looks structured or fluid.

BENDING - *Resistance to folding*

When you bunch up the denim it resists. It springs back toward its original shape, and if you force it to crease the folds are sharp because the fabric is fighting the folding. Think of it as ice. It doesn't bend gradually, it holds and then breaks

sharply. This is why stiff fabrics work for armour, capes and structured collars. They won't just drape, they hold their shape.

The knit offers no resistance at all. It collapses immediately into soft rounded folds with no sharp creases. Low bending resistance, especially on the warp. This fabric goes wherever gravity takes it, producing lots of small soft folds rather than a few large sharp ones. Think of it as water, it flows easily into whatever shape the space allows.

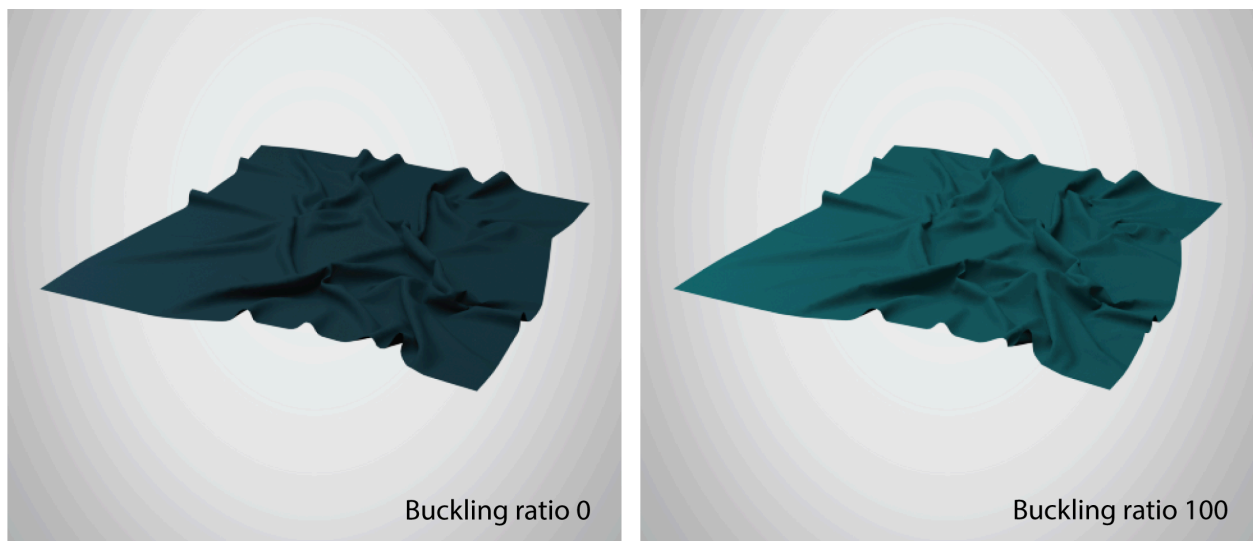


Bending resistance is probably the most visible slider. It's the difference between a cloak that reads as fabric and one that reads as a paper cutout. It also directly affects fold count: stiff fabrics produce fewer larger folds, soft fabrics produce many small ones.

BUCKLING - *What happens when fabric is compressed*

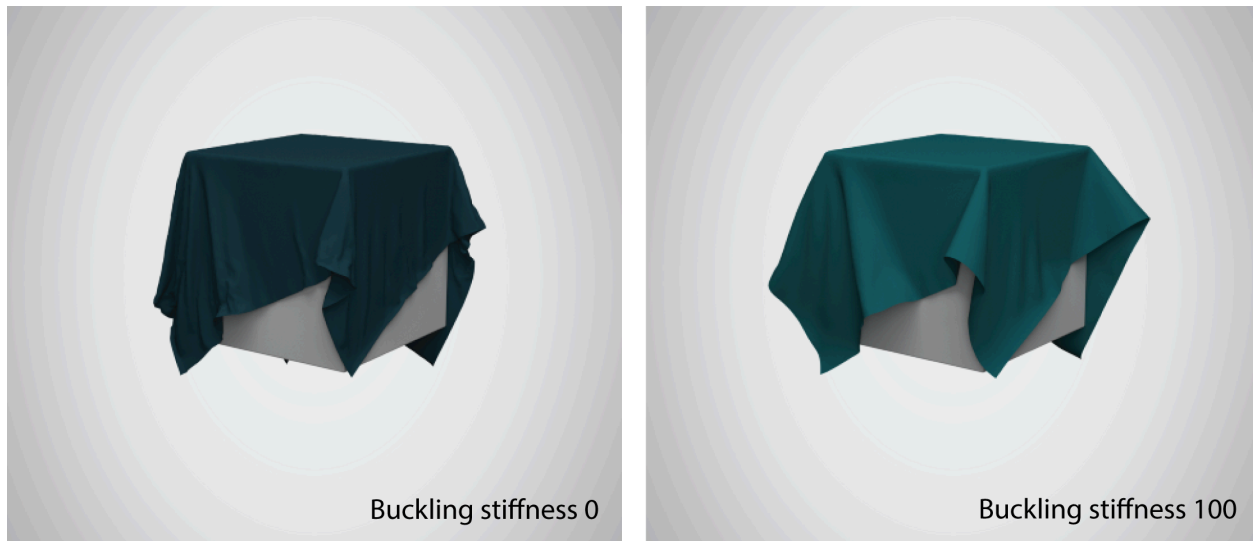
Buckling has two separate controls and it's worth understanding what each one does.

Buckling ratio is about resistance. How much does the fabric fight before it buckles at all? Hold the denim vertically and push the bottom upward. It resists briefly, then buckles into sharp angular folds that stay there. The knit buckles immediately, with almost no resistance before it gives.



Buckling stiffness is about the quality of the crease itself. Once the fabric buckles, is the fold sharp or soft? The denim produces crisp angular folds that hold their

shape. The knit produces rounded gentle folds that partly release when you stop applying pressure. Interestingly the knit's warp stiffness is actually higher than the denim here, which is what stops it from completely collapsing. It buckles but bounces back a little.



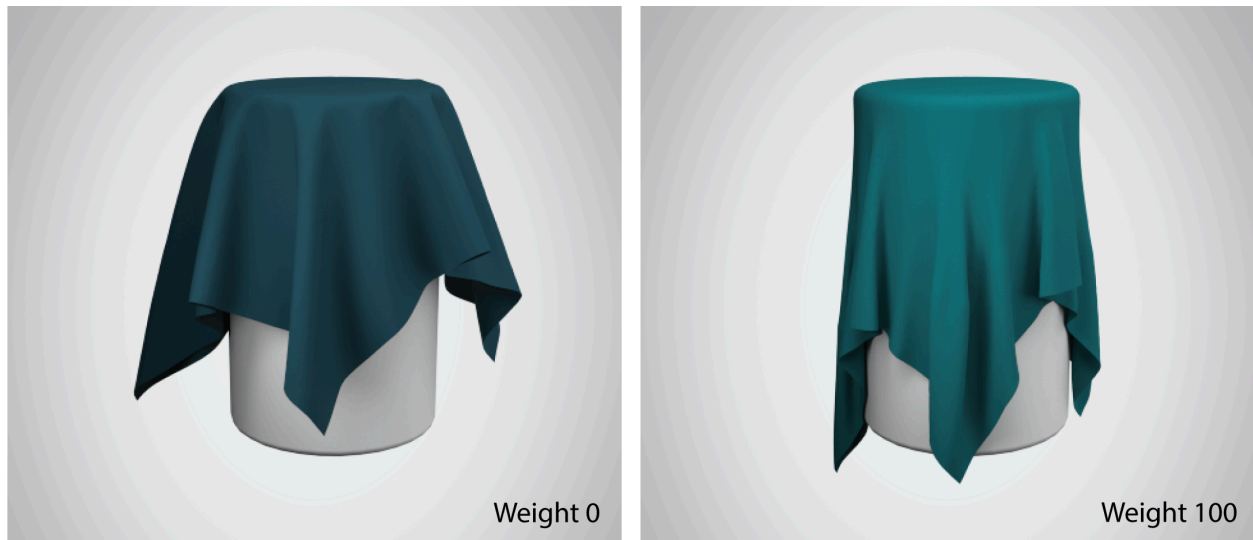
Buckling is what you see at elbows, knees and waists, anywhere fabric gets compressed as a character moves or poses. Getting this wrong is why clothing in games sometimes looks like it's made of rubber or cardboard. Bending and buckling together are what make fabric read as a specific material rather than a generic simulation.

WEIGHT AND INTERNAL DAMPING - *Gravity and momentum*

Hold the denim in the air and let it fall. It drops fast, lands with force and settles almost immediately with minimal movement after. Heavy fabric falls with authority. Low internal damping means it doesn't oscillate much, it just stops. Think of a heavy wool coat when you sit down. It falls and stays.

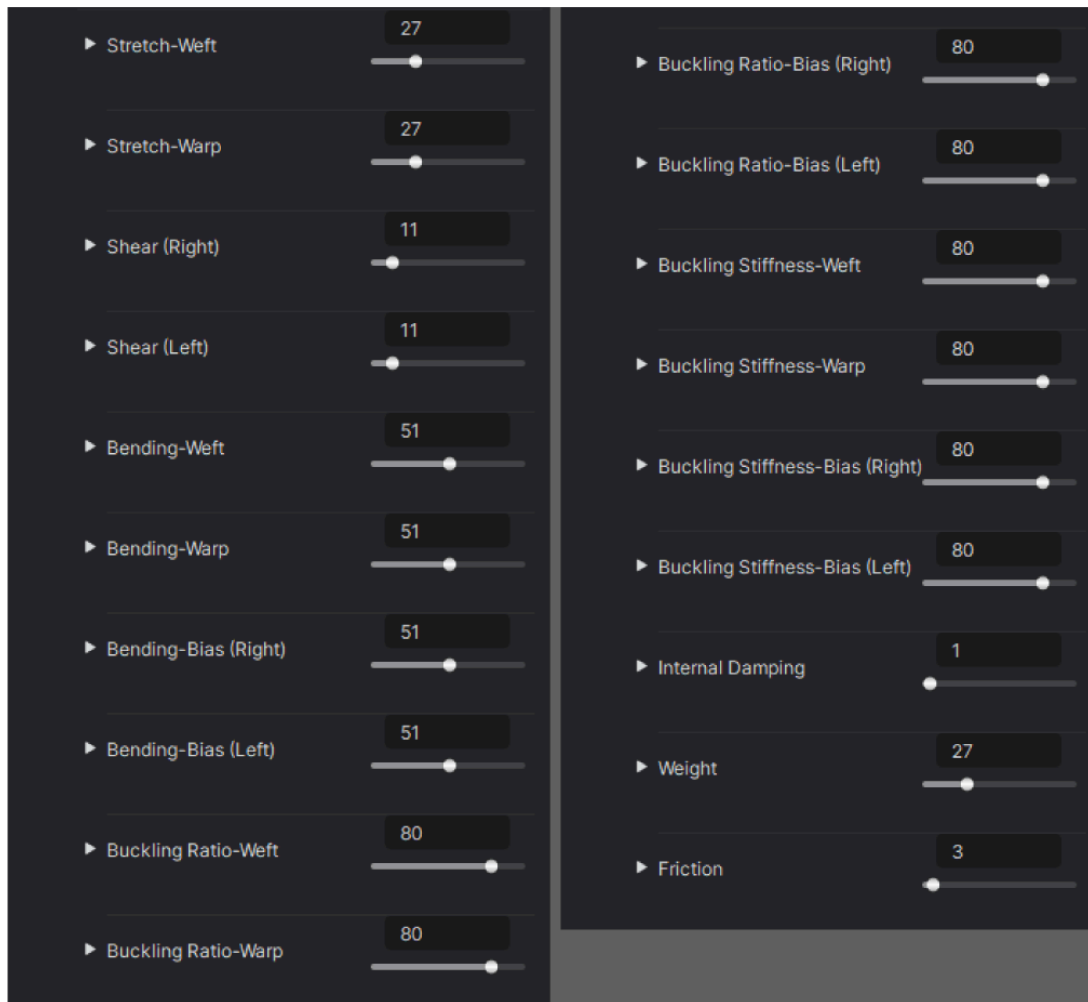
Drape the knit over a finger and it keeps moving after your finger has stopped. Bounce it gently and it wants to oscillate. It's a much lighter fabric, so it responds to even small movements. Even with the same low damping value, the low weight means it keeps moving longer because there's less mass to stop it. Think of a

sheer undershirt on a character who has just stopped running. It keeps swaying slightly.



One thing worth noting about the preset values here: these sliders appears to have used weight itself to tune the settling behaviour, keeping internal damping low for both fabrics. In practice if you were building the knit preset from scratch you would likely want higher internal damping to account for how much it wants to bounce.

Weight and damping together are what make fabric feel alive or dead in animation. A heavy cloak and a light scarf should never move the same way. These sliders are what your technical artist will fine-tune at the end of the pipeline, but understanding them at the concept stage means you're already thinking about how your design will move before it ever gets into the game engine.



^the default sliders used for this tutorial

The sliders in Marvelous Designer are a description of physical behaviour, and if you've ever held a piece of fabric and paid attention to what it was doing, you already have most of the intuition you need to use them well.

Starting from the material means you can make informed decisions before you've run a single simulation. You know your fabric is stiff on the grain and soft on the bias, so you already have a sense of where stretch and shear need to be. The testing and fine-tuning still happens, but now you can iterate from a position of understanding.

If you want to go further, the best next step is to get physical samples of the fabrics you simulate most often and run your own tests. Pull them, bunch them, compress them, drop them. The relationship between what you feel and what the sliders do becomes very intuitive very quickly.

Thanks for reading my tutorial!